

CHAPTER 3

Variables and The Three Claims

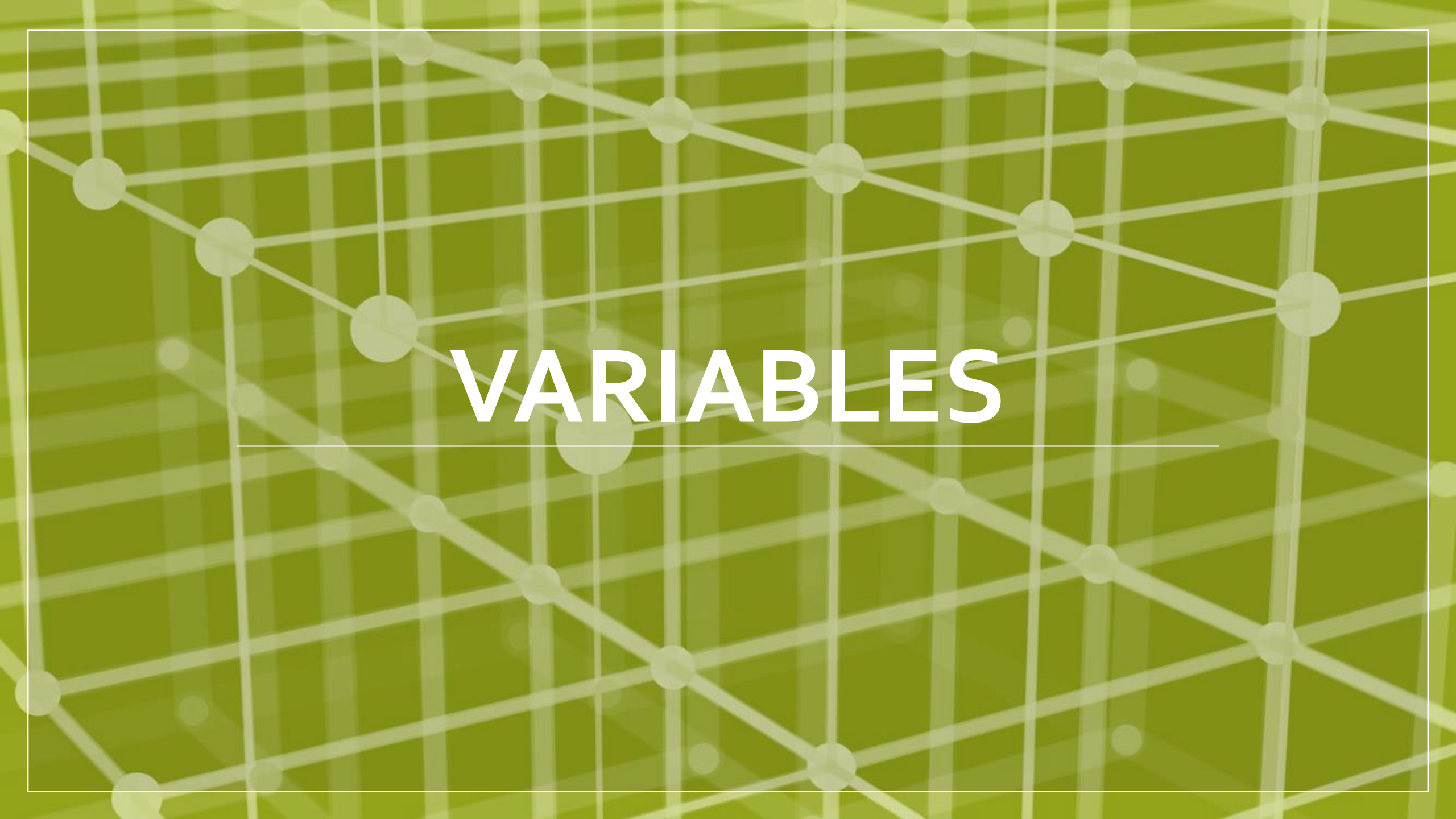
Develop GOOD Research Questions

Ask IMPORTANT QUESTIONS

- justify expense & time
 - Practical
 - Theoretical
 - Advances understanding

Unimportant questions

- Answer firmly established
- Minimal effects on behavior of interest
- No reason to believe variables related



VARIABLES

Variables

- **Variables**: something that varies that we want to measure
- **Conceptual variables/constructs**: abstract names or ideas
- **Operational definition/variables**: a concept of interest that can be measured or manipulated
 - Clearly stated definition
 - Determined by researcher





Operational Definitions - EXAMPLE

- Imagine that I want to study if eating chocolate before bed improves sleep
- How to operationally define chocolate? Sleep?

Attendance Activity

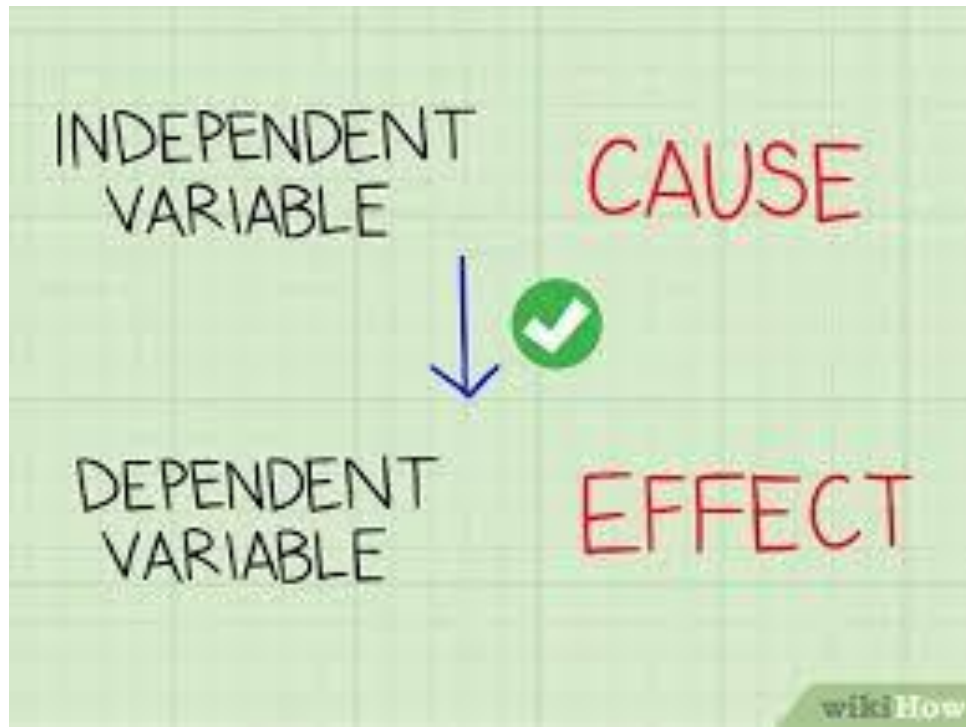
- Now come up with a research question
- What are the constructs in your research question?
- What are the operational definitions of these constructs?

Measured and Manipulated Variables

- **Measured variables**
 - observed
 - recorded
- **Manipulated variables**
 - controlled

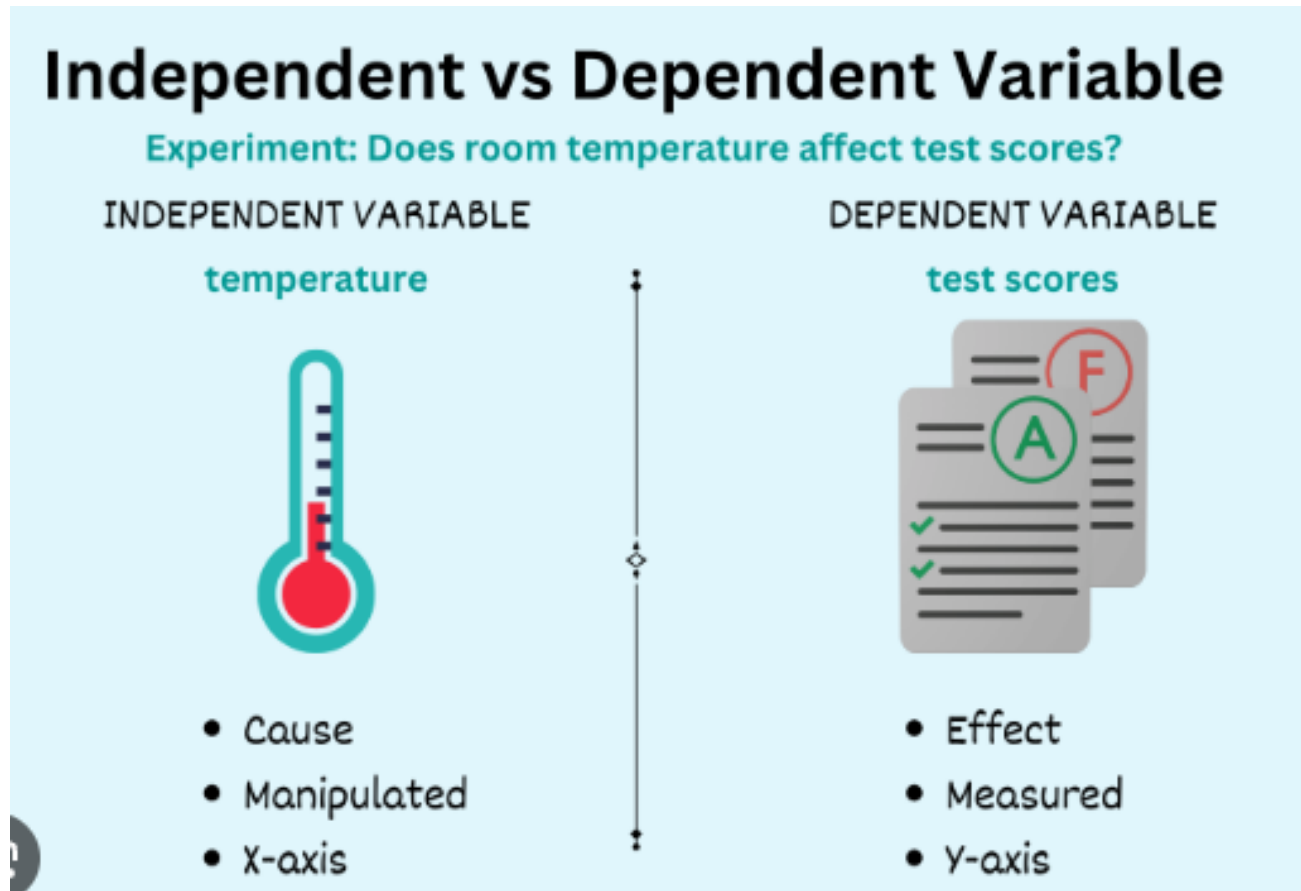


Independent Variables (IV)

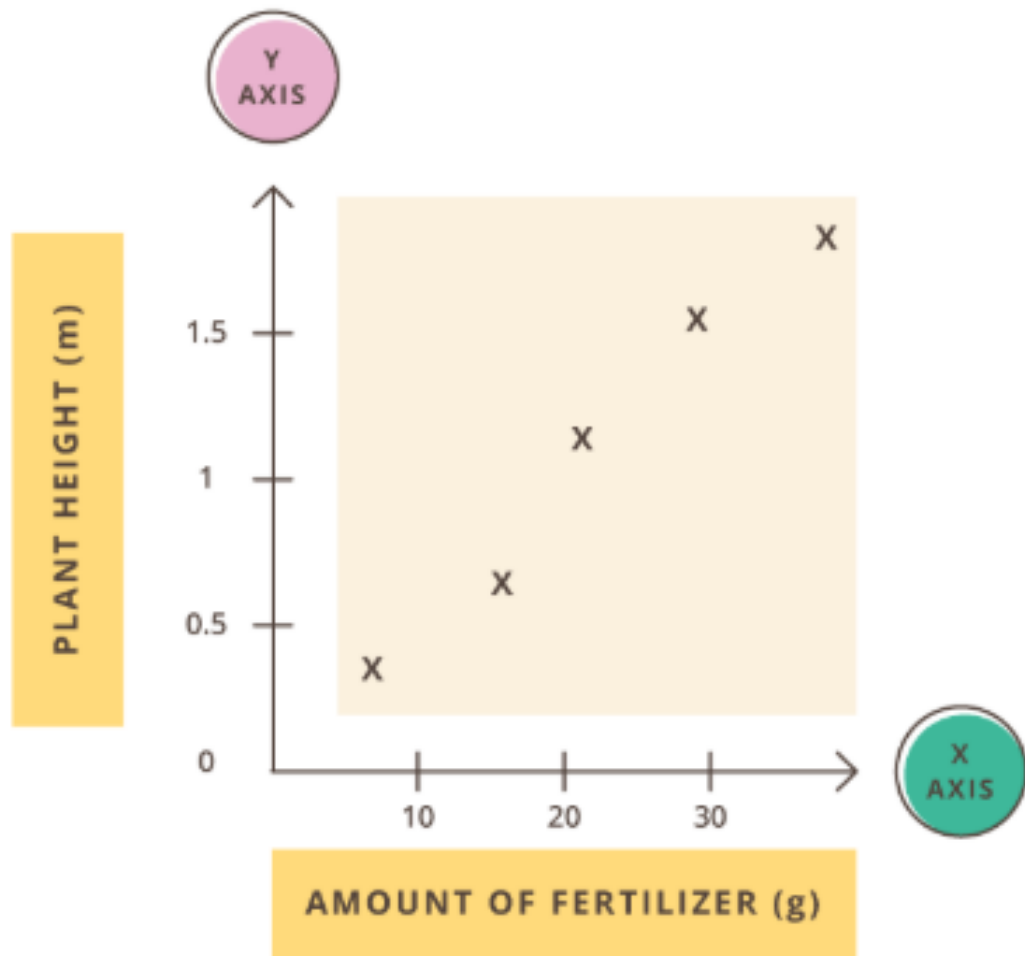


- ***Manipulated***
 - Values determined by researcher
- At least two values
 - Levels
 - Conditions
 - Treatments
 - Groups

Dependent Variables (DV)



- Measured
 - value determined by behavior/response



IV AND DV ON GRAPH

Attendance Activity

- "Students who eat breakfast will perform better on a psychology exam than students who do not eat breakfast."
- IV?
- DV?

Attendance Activity

- “Students who eat breakfast will perform better on a psychology exam than students who do not eat breakfast.”
- IV: breakfast and no breakfast (two levels)
- DV: exam performance



THREE CLAIMS



Claim 1: Frequency Claim

Describes a particular rate or degree of a single variable

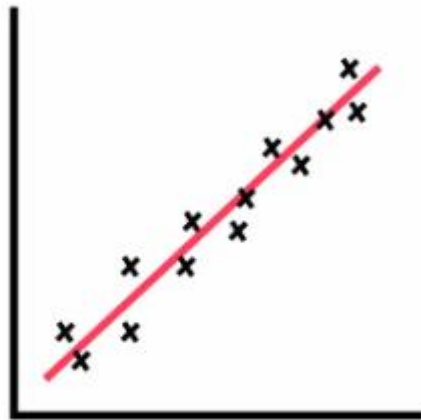
Example: 74% of people smiled yesterday

Claim 2: Association Claim

- Argues that one variable is likely to be *associated* (or correlated) with another variable
- Example: Countries with more butter consumption tend to be happier



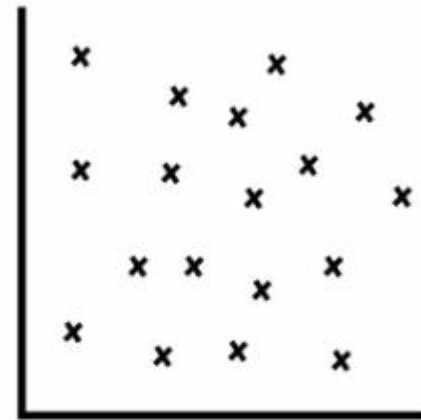
CORRELATIONS



Positive
Correlation

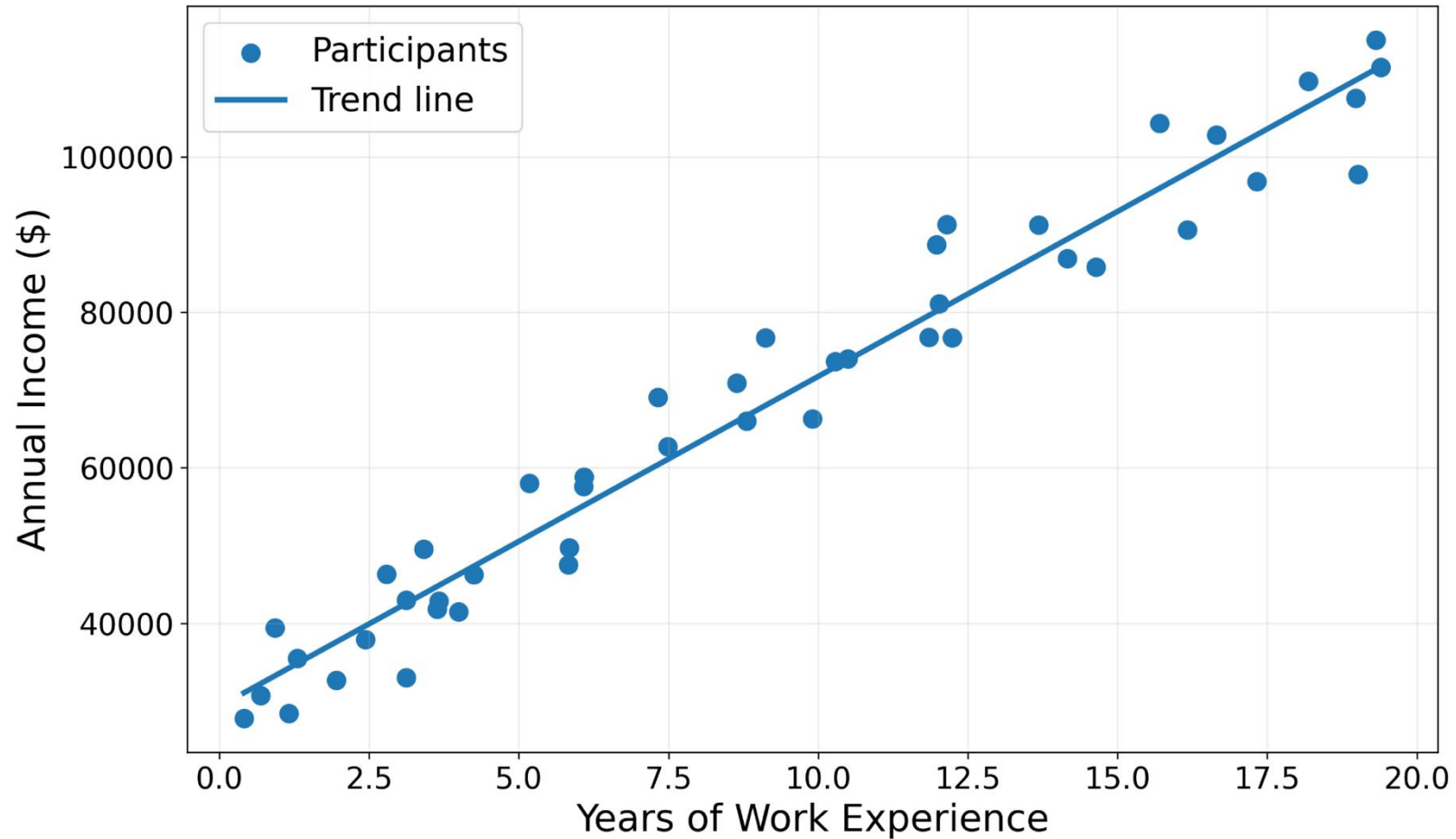


Negative
Correlation

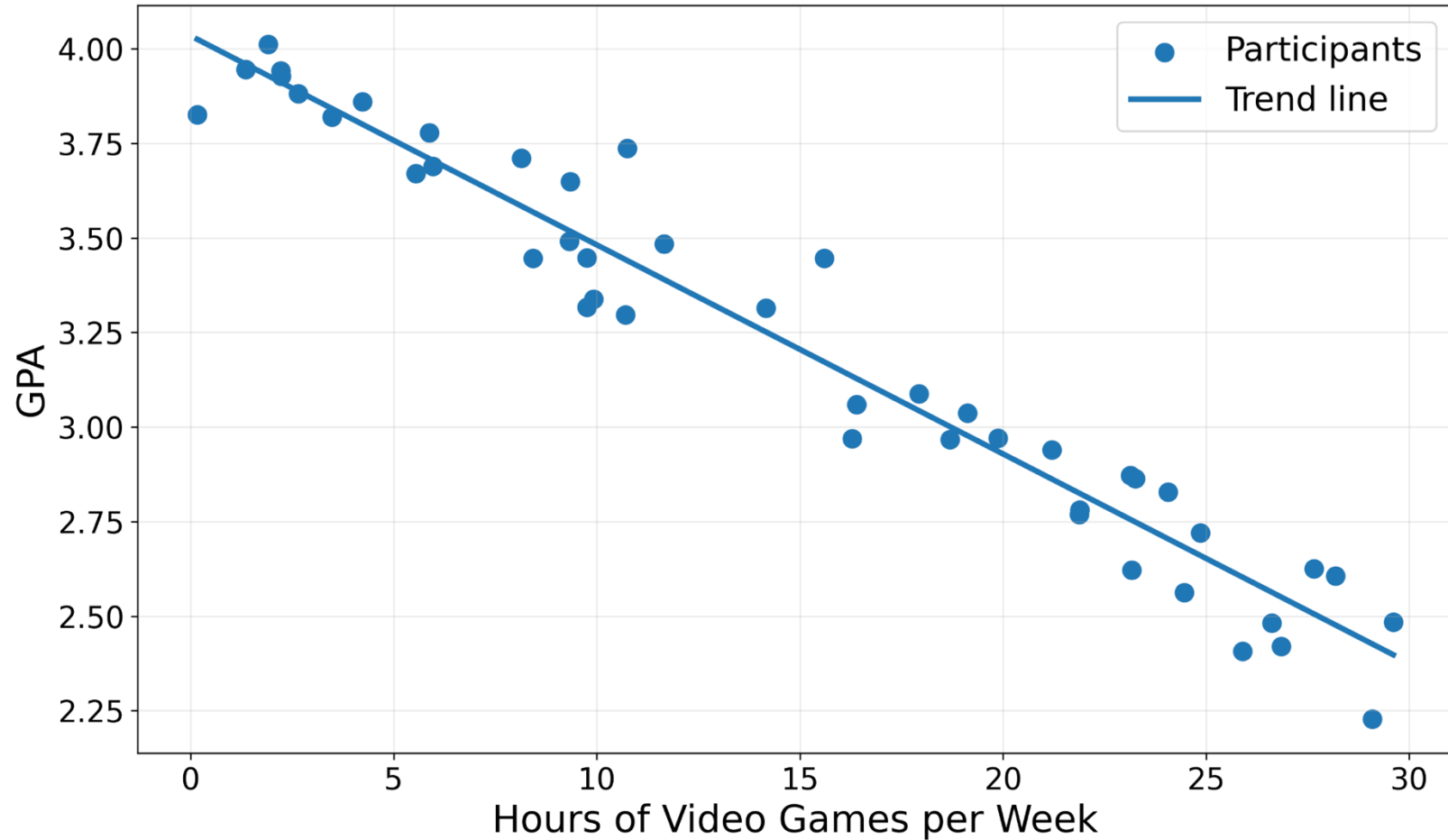


No
Correlation

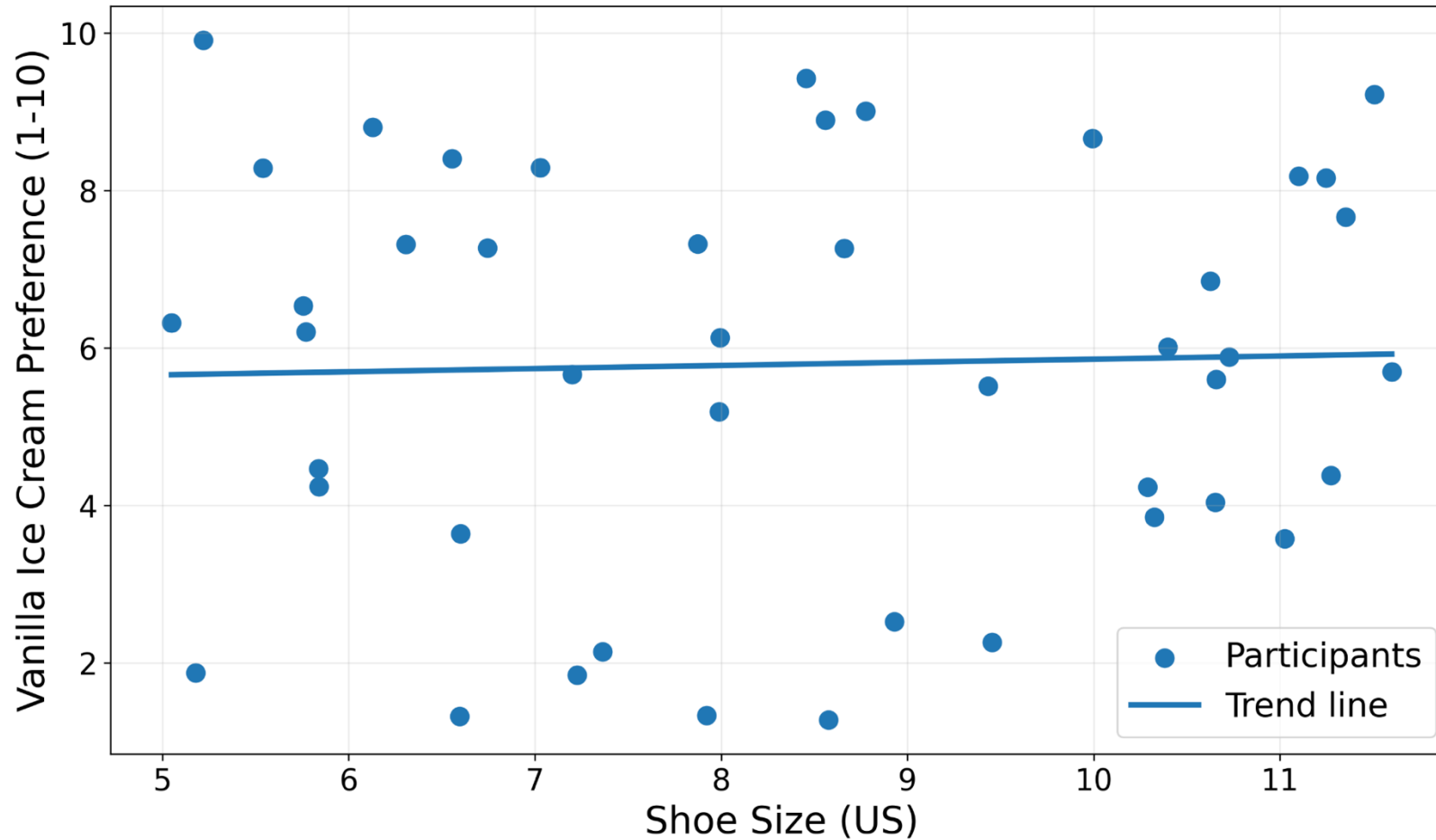
Positive Correlations

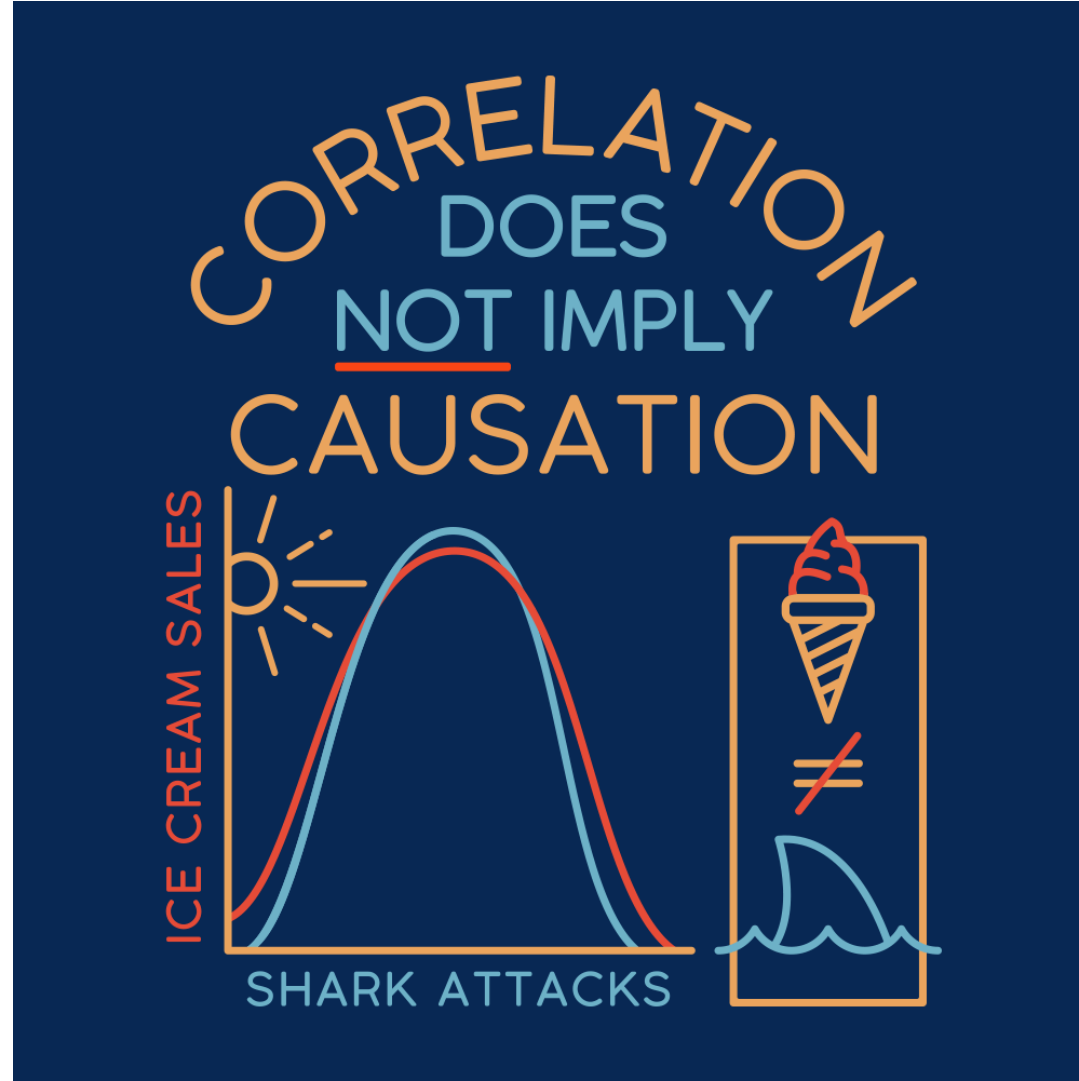


Negative Correlations



No Correlations



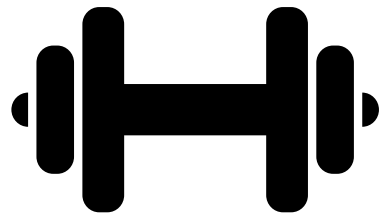


IMPORTANT
NOTE:
**CORRELATION ≠
CAUSATION**

Attendance Activity

- Create one example for each type of correlation.
 - Positive correlation
 - Negative correlation
 - No correlation

Claim 3: Causal Claim



Argues that one variable is responsible for changing the other (e.g., that one *causes* the other to change)

Example: Going to the gym and lifting weights causes you to gain more muscle

Criteria for Causality

1. Covariation: extent to which 2 variables are observed to go together
2. Temporal sequencing: the causal variable comes first/before the effect
3. Internal Validity: ability to eliminate alternative explanations



Demonstrating Causality

- Experiments (studies where one variable is manipulated and the other is measured) can support causal claims
- This is the "gold standard" in research



Be Warned!

A lot of association claims are worded incorrectly in causal language.

Association vs. Casual Claims

- "Social media use is associated with higher anxiety."
- "Social media causes anxiety."
- Which direction?
- Another factor?

Attendance Activity

- “Social media use is associated with higher anxiety.” → “Social media causes anxiety.”
- How to determine the casual relationship?
 - Covariation
 - Temporal Sequencing
 - Internal Validity